

A Novel Visible Light Positioning System Based on the Dual-LED Anchor

Xiaoxiang Cao^{1b}, Xuan Wang^{1b}, Guoliang Chen^{1b}, and Xiansheng Yang^{1b}

Abstract—Recently, indoor localization has played a crucial role in the construction of smart homes, smart buildings, and smart cities. Numerous localization technologies have been investigated, and visible light positioning (VLP) as one of the representative technologies has drawn much attention due to its high accuracy, low cost, and friendly infrastructure. Aiming to solve the prominent issues faced by the traditional VLP ranging-model-based localization method, a phase-difference-based ranging method assisted by a dual-light-emitting diode (LED) anchor was first proposed, which can eliminate the negative influences of hardware heterogeneity, environmental differences, and receiver's inclination angle effectively. The key point of the proposed solution is that we innovatively designed an anchor integrated by two closely spaced LEDs replacing the two separated LEDs for distance calculation, which contributed to avoiding the workload of the ranging model's calibration and the dependence on frequency exchange. Furthermore, a weight matrix was introduced in the final localization to restrain the significant error and increase the confidence coefficient of reliable observations. Finally, a brief analysis of the proposed method's theoretical performance was presented and evaluated by extensive experiments. The proposed method was illustrated to be a very competitive system, which obtained a centimeter-level localization accuracy, and showed more advantages than the traditional ranging method, especially when the photodiode (PD)'s inclination angle is not zero.

Index Terms—Dual-light-emitting diode (LED) anchor, indoor positioning, phase difference, visible light positioning (VLP), weight matrix design.

NOMENCLATURE

f_{10}	Frequency of base LED.
f_{11}	Frequency of localization LED.
θ_{10}	Base LED's radiation angle.
θ_{11}	Localization LED's radiation angle.
ϕ_{10}	PD's incidence angle for base LED.

ϕ_{11}	PD's incidence angle for localization LED.
FOV_{10}	Base LED's field angle.
FOV_{11}	Localization LED's field angle.
A_R	Effective photosensitive reception area of PD.
d_{10}	Distance between PD and the base LED.
d_{11}	Distance between PD and the localization LED.
n	Material reflectance of PD.

I. INTRODUCTION

THE demands for indoor positioning have increased with the development of society and the economy. It is tightly linked with many applications, such as the smart city, smart buildings, and intelligent mines [1]. However, the global navigation satellite system (GNSS) cannot work indoors due to signal obstruction and reflection. Thus, many replaced technologies have been studied [2], and they can be divided into two groups: indoor localization systems (IPS) with base stations and IPS without base stations. The former includes Wi-Fi [3], [4], audio [5], Bluetooth [6], ultrawideband (UWB) [7], visible light positioning (VLP) [8], etc. The latter mainly includes inertial navigation systems (INS) [9], [10], geomagnetic [11], and vision [12]. However, the performance of most technologies is negatively related to the costs, algorithm complexity, resource consumption, etc. [13]. VLP is an excellent solution to these problems [14]. Its high accuracy, lower costs, and friendly infrastructure requirements [14] have drawn much attention recently. Besides, its advantages perfectly meet the requirements of many scenarios, like mines, tunnels, and homes. Thus, this article will mainly discuss the VLP systems based on the multiple anchors, which are widely studied and used.

The VLP systems based on the multiple anchors can be further divided into two groups according to the receiver [14]: systems based on the image sensor (IS) and systems based on the photodiode (PD).

The process of the former method contains two main steps [14]: recognition and calculation. The light-emitting diodes (LEDs) are modulated to different frequencies, and the IS acquires the images of the LEDs. Due to the difference in the frequencies, the images of different LEDs will have stripes of different widths, and this phenomenon is the well-known rolling shutter effect. Thus, it can recognize which LED is observed through the image difference. Then, acquire the observed LEDs' coordinates and calculate the IS's position using the photogrammetry-based method. Much research has been done in this work [15], including encoding methods, image decoding, sensor choice, and localization methods.

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The encoding work uses modulated LEDs to transmit the code sequence for image detection, such as the Manchester encoding [16]. The works of image decoding are equal to image recognition. Extracting the local image of each LED and then doing some work for image enhancement and feature extraction. Recognizing the LED by the intensity of bright and dark bands in the end [17]. As for sensor choice, it means what kind of IS we choose and whether there is an assisted sensor. Because the IS-based method requires an azimuth angle, a single camera obviously cannot meet the localization requirements. Thus, a double-camera-based method is proposed using the proportional relationship of the two triangles to obtain all the distances [18]. Of course, if aided by an external assisted source or sensor to calculate the rotation matrix, a single camera can also work, and the inertial sensors are the first choice [19]. In summary, the IS-based method can obtain high accuracy and is widely used in robots and unmanned vehicles. However, there are still many problems [14]: It is hard to handle the situation if the terminal with a high speed or the image is blocked [20]. The dual-camera-based method has high requirements for the receiver. And the low-cost inertial sensors may lead to an accuracy loss for such a high-accuracy method. Moreover, it is rarely used for pedestrian navigation because it may not meet the user's habits if the application always requires the users to keep their smartphones' cameras turning on.

The PD-based methods can be further divided into several groups according to the observation, such as the received-signal-strength-based (RSS-based) methods, the time-difference-of-arrival-based (TDOA-based) methods [21], and the angle-of-arrival-based (AOA-based) methods [22]. The RSS-based methods have gained much attention recently due to their easy setting, which is widely studied [23]. They can be further divided into two groups: The first is trilateral-based methods [16]. Its basic principle is: first, calculating the distance between the anchor and the receiver by the RSS and an already calibrated ranging model [24]; then, when the observed anchor's number exceeds 3, the receiver's position can be obtained by finding the intersection of multiple circles [14]. Though the principle of this method is simple and understandable, many problems must be solved. The first and foremost is the ranging model, which describes the relationship between the distance and the RSS and is always derived from the signal transmission model [24]. It is originally a complex expression that describes the relationship between light intensity and many factors [14], including the distance, incidence angle, radiation angle, some other hardware parameters, etc. However, It is unrealistic to consider all these factors in each calibration process. Thus, a simplified ranging model is widely used [25]. Just because of this, this method-based VLP system is easily affected by the hardware heterogeneity, receiver's placement, and environmental differences [14]. Significantly, the method we proposed in this article focuses on solving the problems faced by the ranging-model-based method. It eliminates the workload of the model calibration and the influence of different sources. The other is well-studied fingerprint-based methods [26], [27], which contain two main steps [28]: offline fingerprint database construction and online localization.

Offline work collects signal features at many known points, and its research hotspot is the rapid construction of the fingerprint database [29]. Online localization is a neighbor point-matching process with similar signal features. It includes the traditional signal-similarity-distance-based methods [30], the probabilistic-based methods [31], the clustering-based methods [32], and the machine-learning-based [30] or deep-learning-based methods [33]. Nevertheless, fingerprinting-based methods face the main problem which is the workload of the database's construction because of the positive relationship between the density of the fingerprint points and the localization accuracy [33]. This problem limits its application in high-accuracy localization systems like VLP or large scenes.

The TDOA-based method in VLP [21] differs from the traditional TDOA-based method because each LED has a unique frequency. It mainly uses the relationship between the phase difference and distances. However, it requires switching the LEDs' frequencies, which will make anchors unrecognized. Thus, the reasonability of the frequency switching needs to be further verified, and the process of the base distance calculation is too complex. To solve this intractable problem, a dual-LED anchor is designed in this article, and an assumption is used to simplify the distance calculation process. Most AOA-based methods use an elaborate PD array, or LED array [34], which means high requirements for the receiver or the anchor. Besides, a high-complexity localization algorithm is always used in the AOA-based methods, which means high compute resources and limits its deployment on the terminal or the embedded device.

We originally proposed a dual-LED-based VLP system aided by a phase-difference-based ranging method, which entirely eliminates the workload of the ranging model calibration and the influence of PD's inclination angle. The main contributions are as follows.

- 1) To replace the traditional separated LEDs, we proposed a dual-LED anchor-based VLP system for the first time, which dramatically eliminated the workload of the ranging model's calibration and the dependence on frequency exchange.
- 2) To decrease the influence of hardware heterogeneity, environmental differences, and receiver placement, a ranging method based on the phase difference was proposed, which entirely skips the work of the ranging model's calibration and will not be affected by the PD's inclination angle or the change in the environment.
- 3) To restrain the influence of the big error, a weight matrix was used in the final localization. Compared with the method without weight, it gained a better performance and made the trajectory smoother.

The remainder of this article is organized as follows. Section II describes the method we proposed. Section III presents the experiments and results. Finally, the conclusion is drawn in Section IV.

II. METHODOLOGY

In this article, a dual-LED-based VLP system is proposed, and we will briefly describe this system: The anchors are deployed evenly in the scenario, and each anchor integrates two LEDs. These two LEDs are modulated to unique

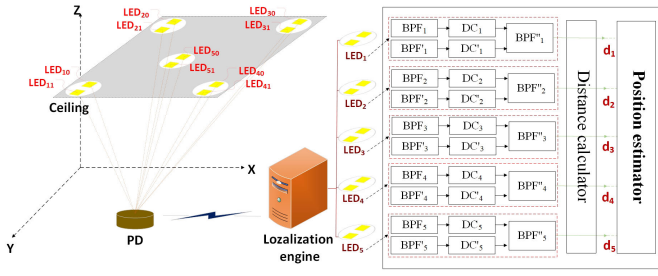


Fig. 1. Components of the proposed VLP system.

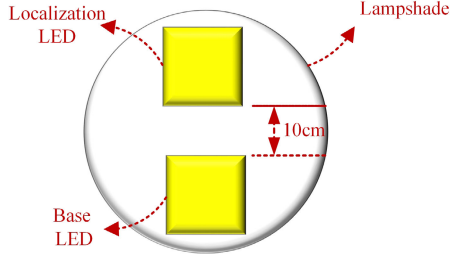


Fig. 2. Layout of the dual-LED anchor.

frequencies, one is regarded as a base frequency and the frequency of the other is the base's odd multiple. Then, the PD receives the signal emitted by the LEDs, and the signal data will be transmitted to the localization engine (LE) using the wireless network to further process because of the terminal's limited calculation and storage source. The LE will use a special-designed bandpass filter (BPF) for each LED's signal separation, and the signals from the one anchor are processed by a down converter (DC) and a BPF. Then a distance can be calculated under the assumed condition that the distances between these two LEDs are approximately equal. Similarly, other distances can be obtained using the same method. Finally, the PD's position can be acquired using the trilateral method assisted by a weight matrix. The whole process of the proposed system is described in detail in Fig. 1.

A. LED Anchor Design

A dual-LED-based anchor is designed for the proposed VLP system, which integrates a base LED and a localization LED. Its layout is shown in Fig. 2. The base LED is just used in distance calculation and will not be used in final positioning. The distances between these two LEDs and PD are almost equal due to the small interval between them. The equality relationship is regarded as an assumption that will be used in the distance calculation. It can avoid using the calibrated ranging model. The design of the anchor has been carefully discussed, mainly the interval of two LEDs. There is a positive relationship between the LEDs' interval and localization performance. However, to guarantee that the phase difference between these two LEDs can be recognized, the interval must be higher than a certain size. Finally, the interval is set to 10 cm by comparing the performance of several simulation experiments, and the detail of the comparison tests will be discussed in Section III.

B. Frequency Design

The frequency design work can be divided into three parts: the first is frequency magnitude selection, the second is

frequency setting in one anchor, and the last is frequency assignment.

1) *Frequency Magnitude Selection*: Because there is a coefficient equal to light speed/(2 · frequency) in the relationship function between the phase difference and distance, it means a tiny error in phase difference may be significantly amplified, and the massive error can further propagate to the distance calculation. Besides, the system may face some problems if the frequency is too high. The first is that it will put severe stress on the hardware and processor's processing capacity, and the second is that it may lead to a large-span frequency range due to the frequency uniqueness of different LEDs and the multiple relationship between two LEDs in one anchor. Thus, the frequency magnitude is set to MHz.

2) *One Anchor's Frequency Setting*: The localization LED's frequency is an odd multiple of the base LED's frequency. This relationship will help us determine the relationship between the distances and the phase difference. This setup also can avoid frequency interference while the even multiple can generate disturbing components. However, if the multiple is too big, it seems unreasonable that there is a large frequency span between two LEDs, such as 1 and 100 MHz. Besides, a large frequency span will cause problems in frequency assignment. Thus, the multiple is set to 3 in this article, and we take one anchor as an example: the base LED's frequency is f_{10} , and the localization LED's frequency will be $f_{11} = 3f_{10}$.

3) *Frequency Assignment*: To avoid frequency interference, the frequency of each LED should be elaborately designed. The process of frequency assignment is a progressive layer of the process. Here, we will take three LEDs as an example.

- 1) *Frequency Assignment for Anchor A (Two LEDs)*: f_{10} and f_{11} .
- 2) *Build an Unavailable Frequency List*: $\text{useless_freq} = [f_{10}, f_{11}, f_{10} - f_{11}, f_{10} + f_{11}, 2f_{10}, 2f_{11}]$.
- 3) *Frequency Assignment for Anchor B (Two LEDs)*: f_{20} and f_{21} . $f_{20}, f_{21} \notin \text{useless_freq}$.
- 4) *Update the Unavailable Frequency List*: $\text{useless_freq} = [f_{10}, f_{11}, f_{10} - f_{11}, f_{10} + f_{11}, 2f_{10}, 2f_{11}, f_{20}, f_{21}, f_{20} - f_{21}, f_{20} + f_{21}, 2f_{20}, 2f_{21}, f_{10} + f_{21}, f_{10} - f_{21}, f_{10} + f_{20}, f_{10} - f_{20}, f_{11} + f_{20}, f_{11} - f_{20}, f_{11} + f_{21}, f_{11} - f_{21}]$.
- 5) *Frequency Assignment for Anchor C (Two LEDs)*: f_{30} and f_{31} . $f_{30}, f_{31} \notin \text{useless_freq}$.
- 6) Update the unavailable frequency list like 4).

When the radiation or incidence angle is up to a threshold, the PD will not receive the LED's signal. Thus, the two very far apart LEDs can be set to the same frequency. In a big scene test, a Bluetooth module in the terminal and anchor can be used to recognize the LEDs which have the same frequency.

C. Ranging Method Based on the Phase Difference

Traditional RSS-based VLP methods are always using a calibrated ranging model, and the complete model describes the relationship between RSS and many factors, such as

the distance, LED's radiation angle, PD's incidence angle, and PD's inclination angle. However, too many independent variables will increase the difficulty of distance calculation and require more observation. Some efforts are made to avoid the effects of different factors, such as careful device deployment, and further simplify the model. One of the widely used simplified models just describes the relationship between distance and RSS under the assumption that the receiver is always horizontal, as shown in (1) [25]. Nevertheless, the PD's inclination angle is time-varying, which is an essential factor in ranging accuracy. Thus, ignoring or regarding it as a constant is unreasonable, especially in pedestrian navigation

$$d = a \cdot \text{RSS}^b \quad (1)$$

where a and b are the parameters of the simplified ranging model; d represents the distance.

To solve the issue that the traditional calibrated-model-based ranging methods face, a phase-difference-based method is proposed. A dual-LED anchor is an essential precondition for this proposed method. There are some common variables defined, and their abbreviations are shown in Nomenclature. One anchor is taken as an example in the following description. The proposed method contains four steps: signal modulation, transmission, reception, and processing. A detailed description is as follows.

1) *Signal Modulation*: Two LEDs in one anchor are modulated to unique frequencies [21]. Thus, the modulated signals of these two LEDs can be described as follows:

$$\begin{cases} P_{\text{LED}}^{10}(t) = P_C + P_M \cos(2\pi f_{10}t + \varphi_0) \\ P_{\text{LED}}^{11}(t) = P_C + P_M \cos(2\pi f_{11}t + \varphi_0) \end{cases} \quad (2)$$

where P_C and P_M are the direct-current and alternating-current components of the light signal; φ_0 is the initial phase.

Simulation experiments are performed to evaluate the proposed method. Thus, several important errors are added to the signals to close the real test. The first is the time synchronization error, which describes the time difference between two LEDs, and the following equation is achieved:

$$\begin{cases} P_{\text{LED}}^{10}(t) = P_C + P_M \cos(2\pi f_{10}t + \varphi_0) \\ P_{\text{LED}}^{11}(t) = P_C + P_M \cos(2\pi f_{11}(t + \Delta t_{10-11}) + \varphi_0) \end{cases} \quad (3)$$

where Δt_{10-11} represents the time difference between the base LED and localization LED, and the time system of the base LED is regarded as a base.

Although a wired connection means a tiny time synchronization error, which can be less than 1 ps, we still add a random error in the range of 1 ps–1 ns.

2) *Signal Transmission*: The visible light signals emitted by LEDs are transmitted in free space and obey the Lambert emission law. We set the channel gain to $h(t)$

$$\begin{cases} h_{10}(t) = \frac{n+1}{2\pi} \cos^n(\phi_{10}) \cos(\theta_{10}) \frac{A_R}{d_{10}^2} R(\xi_{10}) \delta(\varsigma_{10}) \\ h_{11}(t) = \frac{n+1}{2\pi} \cos^n(\phi_{11}) \cos(\theta_{11}) \frac{A_R}{d_{11}^2} R(\xi_{11}) \delta(\varsigma_{11}) \end{cases} \quad (4)$$

where $\xi_{10} = (\theta_{10}/\text{FOV}_{10})$, $\xi_{11} = (\theta_{11}/\text{FOV}_{11})$, $\varsigma_{10} = t - (d_{10}/c)$, and $\varsigma_{11} = t - (d_{11}/c)$; $R(\cdot)$ is a two-valued function, and they will output 1 if the input's absolute value

is less than 1; otherwise, it is 0. $\delta(\cdot)$ is the channel impulse response function.

3) *Signal Reception*: PD receives a synthetic signal. Except for the signal component with f_{10} and f_{11} , there are some components with new frequencies. The received signal can be described as follows:

$$E_{\text{PD}}(t) = k \cdot |r \cdot [P_{\text{LED}}^{10} \otimes h_{10}(t) + P_{\text{LED}}^{11} \otimes h_{11}(t)]|^2 \quad (5)$$

where $\otimes$ represents the convolution operator; k is a scaling factor; r is the response ratio of the PD.

Then, another error called receiver noise is added to the signals. It mainly contains two parts: shot noise and thermal noise [35]. And the shot noise is affected by ambient noise and signal light disturbance. Considering the noise contains several complex components, we combine these noises and regard them as a random error. The magnitude of the noise is set to 1/10 of the signal's magnitude, and the proportionality is obtained in the real tests. Thus, (5) can be described as follows:

$$E_{\text{PD}}(t) = k \cdot |r \cdot [(P_{\text{LED}}^{10} \otimes h_{10}(t) + e_{10}) + (P_{\text{LED}}^{11} \otimes h_{11}(t) + e_{11})]|^2 \quad (6)$$

where e_{10} and e_{11} are receiver noises.

The following contents are the description of signal processing.

4) *Signal Separation*: The received signal is a synthetic signal, and it integrates many components with different frequencies. Thus, it is necessary to separate each LED's signal first, and two BPFs are used to process the signals. The filter's frequency pass width is f_{width} . The cutoff frequency of BPF for the base LED's signal separation is $[f_{10} - f_{\text{width}}/2, f_{10} + f_{\text{width}}/2]$, and the other is $[f_{11} - f_{\text{width}}/2, f_{11} + f_{\text{width}}/2]$. The signals processed by the BPFs are shown as follows:

$$\begin{cases} E_{10}(t) = L_{10} \cdot \cos[2\pi f_{10}(t - d_{10}/c) + \varphi_0] \\ E_{11}(t) = L_{11} \cdot \cos[2\pi f_{11}(t - d_{11}/c) + \varphi_0] \end{cases} \quad (7)$$

where L_{10} and L_{11} are the magnitude of the separated signals; c represents the light speed. Their complete expressions are as follows:

$$\begin{cases} L_{10} = 2kr^2 (H(0)_{10}^2 + H(0)_{10}H(0)_{11}P_C P_M \\ \quad + 2P_M H(0)_{10}(e_{10} + e_{11})) \\ L_{11} = 2kr^2 (H(0)_{11}^2 + H(0)_{10}H(0)_{11}P_C P_M \\ \quad + 2P_M H(0)_{11}(e_{10} + e_{11})) \end{cases} \quad (8)$$

where $H(0)_{10} = (n+1/2\pi)\cos^n(\psi_{10})\cos(\theta_{10})(A_R/d_{10}^2)R(\xi_{10})$ and $H(0)_{11} = (n+1/2\pi)\cos^n(\psi_{11})\cos(\theta_{11})(A_R/d_{11}^2)R(\xi_{11})$.

It is obvious that the signal's magnitude is related to the radiation angle and incidence angle. This means the widely used simplified ranging model will not be accurate enough if these factors are not taken into consideration.

5) *Signal Group*: In the examples given throughout this article, there are only two LEDs. Thus, the signals that come from the same anchor are easily grouped. However, in the real test, there will be many signals with different frequencies after being processed by the BPF. It means that we should divide the signals that come from the same anchor into a group for the further process.

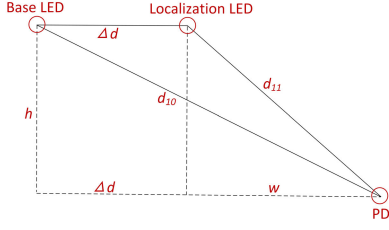


Fig. 3. Relationship between LED interval and the distance between LED and PD.

6) *Frequency Unification*: The signals in (7) are with different frequencies. However, phase difference calculation requires the signals to be with a unified frequency. Thus, a DC and a BPF are designed to process the signals, and their frequencies are changed to $f_{10}/2$. The following equation is achieved:

$$\begin{cases} E'_{10}(t) = L'_{10} \cos\left\{\pi f_{10}t - \frac{2\pi f_{10}d_{10}}{c} + \varphi'\right\} \\ E'_{11}(t) = L'_{11} \cos\left\{\pi f_{10}t - \frac{6\pi f_{10}d_{11}}{c} + \varphi'\right\} \end{cases} \quad (9)$$

where L'_{10} and L'_{11} are the magnitude of the signal; φ' is the total phase shift.

7) *Phase Difference Calculation Using Phase*: After frequency unification, the phase difference between these two signals can be obtained by calculating the difference between their phase

$$\Delta\varphi_{11-10} = \frac{2\pi f_{10}d_{10}}{c} - \frac{6\pi f_{10}d_{11}}{c} = \frac{2\pi f_{10}}{c}(d_{10} - 3d_{11}). \quad (10)$$

8) *Phase Difference Calculation Using IQ*: There is another method for phase difference calculation. It uses the signals and their Hilbert transformation is called the IQ-based method. Thus, we should calculate the IQ values first, and the following equations can be used:

$$\begin{cases} I_{11-10} = E'_{10}(t) \cdot \text{Hilb}[E'_{11}(t)] - \text{Hilb}[E'_{10}(t)] \cdot E'_{11}(t) \\ Q_{11-10} = E'_{10}(t) \cdot E'_{11}(t) + \text{Hilb}[E'_{10}(t)] \cdot \text{Hilb}[E'_{11}(t)] \end{cases} \quad (11)$$

where $\text{Hilb}[\cdot]$ represents the Hilbert transform function.

Then, calculating the phase difference using the IQ values can be described as follows:

$$\Delta\varphi_{11-10} = \tan^{-1}\left(\frac{I_{11-10}}{Q_{11-10}}\right). \quad (12)$$

9) *Ranging*: The relationship between (10) and (12) is used to calculate the distance. Combining these two equations results in the following equation:

$$\Delta\varphi_{11-10} = \frac{2\pi f_{10}}{c}(d_{10} - 3d_{11}) = \tan^{-1}\left(\frac{I_{11-10}}{Q_{11-10}}\right). \quad (13)$$

It is impossible to obtain the distance through (13) because there are two independent variables in it. Thus, an assumption is used to simplify the equation. Considering the interval between two LEDs is far smaller than the distance between LED and PD, there are reasons to believe that $d_{10} \approx d_{11}$. The rationality of this assumption can be explained using Fig. 3.

The simplified equation is as follows:

$$\Delta = d_{11} - d_{10} = \frac{\Delta d^2 + 2w\Delta d}{d_{11} + d_{10}} \quad (14)$$

where Δd is the LED interval; w represents the LED's cover radius; Δd^2 is very tiny. Similarly, $2w\Delta d/d_{10} + d_{11}$ is very small. Naturally, the value of Δ is also very small.

Then, (13) can be further simplified

$$-\frac{4\pi f_{10}}{c}d_{11} = \tan^{-1}\left(\frac{I_{11-10}}{Q_{11-10}}\right). \quad (15)$$

Finally, a distance can be calculated using the following equation:

$$d_{10} \approx d_{11} = -\frac{c}{4\pi f_{10}} \tan^{-1}\left(\frac{I_{11-10}}{Q_{11-10}}\right). \quad (16)$$

Similarly, the distances of other anchors can be calculated using the same method.

In the end, we will briefly analyze why the PD's inclination angle will not affect distance accuracy if using the phase-difference-based ranging method. It is clear that the PD's inclination angle directly affects PD's incidence angle, and the incidence angle further impacts the signal's amplitude. However, the BPF, DC, or Hilbert transformation will not change the signal's amplitude. When we calculate the value of the I_{11-10}/Q_{11-10} , the signal's amplitude is going to cancel. Naturally, the incidence angle is also going to cancel, as shown in the following equation:

$$\frac{I_{10-11}}{Q_{10-11}} = \frac{L_{10} \cdot L_{11}}{L_{10} \cdot L_{11}} \cdot \frac{(E''_{10}(t) \cdot \text{Hilb}[E''_{11}(t)] - \text{Hilb}[E''_{10}(t)] \cdot E''_{11}(t))}{(E''_{10}(t) \cdot E''_{11}(t) + \text{Hilb}[E''_{10}(t)] \cdot \text{Hilb}[E''_{11}(t)])} \quad (17)$$

where $E'_{10}(t) = L_{10}E''_{10}(t)$, $E'_{11}(t) = L_{11}E''_{11}(t)$; $E''_{10}(t)$ and $E''_{11}(t)$ are the cosine functions; L_{10} and L_{11} are related to the incidence angle, noises, and many other hardware-related parameters.

D. Weighted-Least-Squares-Based Localization Method

As long as the number of distances is more than 3, the trilateral-based method can be used to obtain the receiver's position. Here we take five anchors as an example. The relationship between distances and the anchors' positions is shown in the following equation:

$$\begin{cases} (x_{11} - x)^2 + (y_{11} - y)^2 + (z_{11} - z)^2 = d_{11}^2 & (a) \\ (x_{21} - x)^2 + (y_{21} - y)^2 + (z_{21} - z)^2 = d_{21}^2 & (b) \\ (x_{31} - x)^2 + (y_{31} - y)^2 + (z_{31} - z)^2 = d_{31}^2 & (c) \\ (x_{41} - x)^2 + (y_{41} - y)^2 + (z_{41} - z)^2 = d_{41}^2 & (d) \\ (x_{51} - x)^2 + (y_{51} - y)^2 + (z_{51} - z)^2 = d_{51}^2 & (e) \end{cases} \quad (18)$$

where (x, y, z) is the coordinate of the PD; (x_{i1}, y_{i1}, z_{i1}) represents the coordinate of i_{th} LED ($i = 1, 2, 3, 4, 5$).

In (17), PD's 3-D coordinates can be directly obtained using the Yacobi, Newton-Raphson, or other methods. However, multiple iterations and nonlinear mean a high computational

load if these methods are used, which will bring difficulties if the system is deployed in an embedded platform, like a microcontroller. Thus, two assumptions are used to relieve the computational load further: 1) all the anchors are always deployed on the ceiling, meaning they have the same height ($z_{11} \approx z_{21} \approx z_{31} \approx z_{41} \approx z_{51}$) and 2) this system is IPS-application-oriented, and the PD is always attached to a robot or intelligent vehicle; thus, z can be regarded as a fixed value. To sum up, we can assume that $(z_{11} - z) \approx (z_{21} - z) \approx (z_{31} - z) \approx (z_{41} - z) \approx (z_{51} - z)$, and an internal subtraction can be done in (17), which can simplify it to the maximum extent. Equation (17) and (b) – (a), (c) – (a), (d) – (a), and (e) – (a) in (18) can be simplified as follows:

$$A \begin{bmatrix} x & y \end{bmatrix}^T = B \quad (19)$$

where

$$A = 2 \begin{bmatrix} x_{11} - x_{21} & y_{11} - y_{21} \\ x_{11} - x_{31} & y_{11} - y_{31} \\ x_{11} - x_{41} & y_{11} - y_{41} \\ x_{11} - x_{51} & y_{11} - y_{51} \end{bmatrix}$$

and

$$B = \begin{bmatrix} d_{21}^2 + x_{21}^2 + y_{21}^2 - (d_{11}^2 + x_{11}^2 + y_{11}^2) \\ d_{31}^2 + x_{31}^2 + y_{31}^2 - (d_{11}^2 + x_{11}^2 + y_{11}^2) \\ d_{41}^2 + x_{41}^2 + y_{41}^2 - (d_{11}^2 + x_{11}^2 + y_{11}^2) \\ d_{51}^2 + x_{51}^2 + y_{51}^2 - (d_{11}^2 + x_{11}^2 + y_{11}^2) \end{bmatrix}.$$

The receiver's position is obtained by the least-square-based method [36] using the following equation:

$$\begin{bmatrix} x \\ y \end{bmatrix} = (A^T A)^{-1} \cdot A^T B. \quad (20)$$

However, an assumption is used in distance calculation which is described above, and it is clear that the ranging accuracy is related to the distance between PD and LED under this assumption. A larger distance means a higher ranging accuracy. Thus, we use the distance to build a weight matrix W directly, which will be used in (19)

$$\begin{bmatrix} x \\ y \end{bmatrix} = (A^T W^{-1} A)^{-1} \cdot A^T W^{-1} B \quad (21)$$

where

$$W = \begin{bmatrix} d_{21} & 0 & 0 & 0 \\ 0 & d_{31} & 0 & 0 \\ 0 & 0 & d_{41} & 0 \\ 0 & 0 & 0 & d_{51} \end{bmatrix}.$$

III. EXPERIMENTS AND ANALYSIS

A set of experiments are performed to evaluate the proposed method, including actual and simulation tests. Actual tests are used to reveal the problems we face when we use traditional methods. Due to the limitations of the hardware capacity, the method evaluation uses simulation tests rather than real tests. Furthermore, many comparison tests are performed to give us an intuitive understanding of the proposed method and help us analyze the performance of different methods.

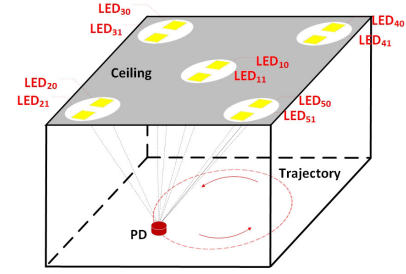


Fig. 4. Setup of the experiment scene and deployments of the anchors.

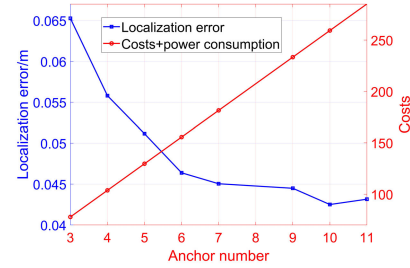


Fig. 5. Relationship between anchor number and localization error.

A. Simulation Test Setting

There is a $10 \times 10 \times 3$ m room, and five dual-LED anchors are deployed in this room, including one LED in the center of the room and the rest in the corners. The distribution is shown in Fig. 4, and the distribution design takes the same approach as most RF-based IPSs [7], [37]. Moreover, we evaluated the performance of the different anchor numbers, i.e., 3, 4, 5, 6, 7, 9, 10, and 11 anchors were chosen to evaluate. Some points are chosen to test, and they are evenly distributed in the whole simulated area, and the space between them was fixed (0.5×0.5 m). Fig. 5 shows the mean localization error of different anchor numbers, and it is clear that more anchor means higher accuracy. However, the accuracy difference is insignificant; for example, there is only a 1.0 cm difference when using 5 and 11 anchors. Moreover, the costs (the price of each LED bead is about \$0.15) and the power consumption (10 W per LED) of different anchor numbers also were calculated. The results are shown in Fig. 5, which indicated that choosing five anchors was finally proved to be the optimal result after balancing localization accuracy, costs, and power consumption due to the costs curve and localization error curve intersecting near five anchors.

The field of view (FOV) angle of the LED and PD is set to 75° . All the LEDs are deployed on the ceiling. Thus, all of them are always regarded as horizontal. Besides, if the PD is also horizontal, the maximum coverage area of a single anchor can be a circle of radius 11 m. However, if the acceptance angle is larger than half of the PD's FOV angle, the PD will not receive the signal. More precisely, the received signal's amplitude is the same as the noise (nearly 1/100 of the signal).

The frequencies of these anchors are set to $(f_{10}, f_{11}) = (1.0 \text{ MHz}, 3.0 \text{ MHz})$, $(f_{20}, f_{21}) = (1.3 \text{ MHz}, 3.9 \text{ MHz})$, $(f_{30}, f_{31}) = (1.9 \text{ MHz}, 5.7 \text{ MHz})$, $(f_{40}, f_{41}) = (2.1 \text{ MHz}, 6.3 \text{ MHz})$, and $(f_{50}, f_{51}) = (2.5 \text{ MHz}, 7.5 \text{ MHz})$.

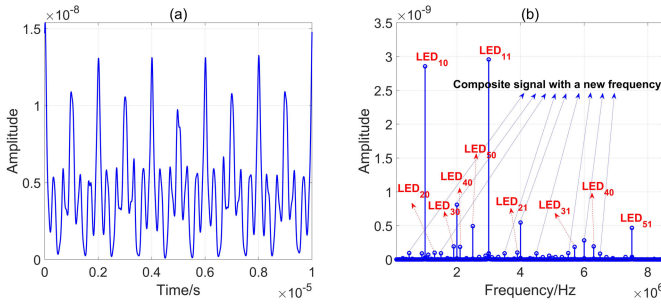


Fig. 6. Received signal and its FFT results. (a) Received signal. (b) FFT results of the received signal.

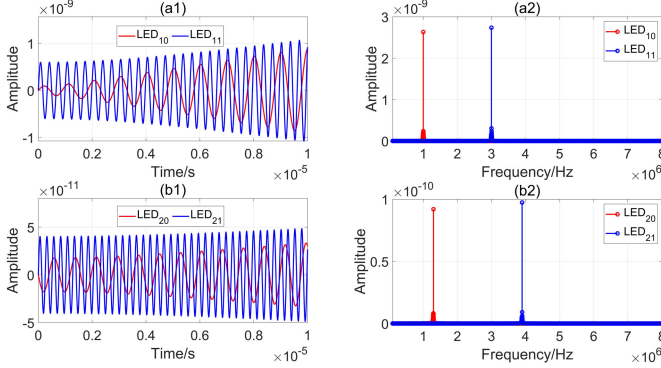


Fig. 7. Signal processed by a BPF and their FFT results. (a1) Signals of anchor A. (a2) FFT results of anchor A's signals. (b1) Signals of anchor B. (b2) FFT results of anchor B's signals.

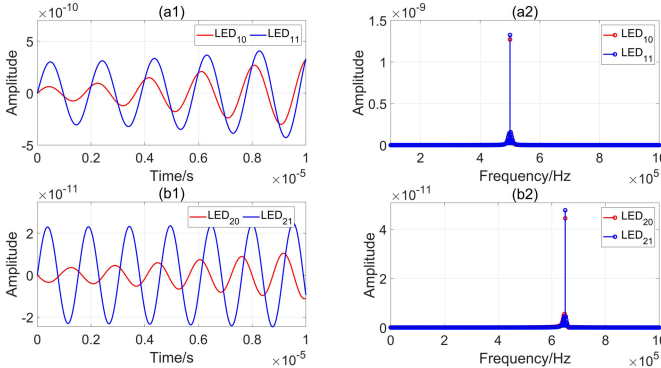


Fig. 8. Signal processed by a DC and a BPF, and their FFT results. (a1) Signals of anchor A. (a2) FFT results of anchor A's signals. (b1) Signals of anchor B. (b2) FFT results of anchor B's signals.

B. Signal Simulation

The complete signal transmission process is simulated, from modulation to reception and processing. Several errors are added to get closer to reality, mainly including the time synchronization error and receiver noise.

Fig. 6 shows the received signal with added errors and its fast Fourier transformation (FFT). It is clear that many components with new frequencies in the received signal, except the frequencies we modulated, are generated. These interference components with new frequencies can be eliminated using elaborate-designed BPFs, and each LED's signal can also be separated from the raw signal for the following processing. What needs to be emphasized is that the amplitude of the time synchronization error is [1 ps, 1 ns], and the receiver noise's amplitude is 1%–10% of the raw signal's amplitude (which is obtained in the actual tests). It is certain

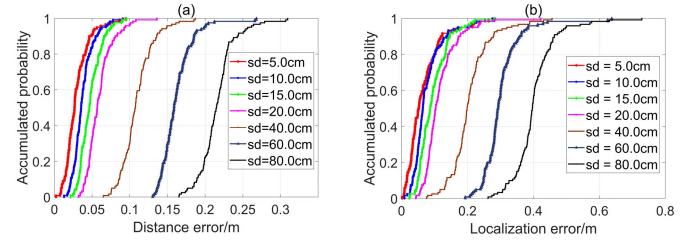


Fig. 9. Performance of different LED intervals. (a) CDF of the distance error. (b) CDF of the localization error.

that the time synchronization error we added is larger than the reality because the wired synchronization system always causes a very tiny time difference between different LEDs which means a less than 1-ps-level time synchronization error. Fig. 7 shows the separated signals and their FFTs. Here two anchors' signals are shown. Then, a DC and a BPF are used to unify the frequencies. The signals processed by the DC and BPF are shown in Fig. 8. We can find that the frequencies of two LEDs in one anchor are unified to half of the base LED's frequency.

C. LED's Interval Analysis

A dual-LED anchor is designed for the proposed method. Thus, the first job is to decide the optimal interval of these two LEDs. Several intervals are selected to test their performance (from 5 to 80 cm), and the results are shown in Fig. 9. Mainly including the cumulative distribution function (cdf) of distance and localization errors (441 points are selected for evaluation, which is evenly distributed in the whole localization area). It is obvious that a smaller interval means better performance. However, a tiny size may cause some problems, including the lampshade's size decreasing the FOV of the LED and lowering the distinguishability of the phase difference. Thus, the interval is set to 10 cm in this article, which has most of the same performance as 5 cm, and nearly 95% of localization errors are less than 10 cm.

D. Theoretical Performance Analysis

Some indicators are used to reflect the theoretical performance of the localization algorithm. The Cramer–Rao lower bound (CRLB) [38] and geometric dilution of precision (GDOP) [39] are selected to evaluate the proposed method's theoretical performance in this article. CRLB value is tightly linked with the mean square error of the localization algorithm, and the GDOP value can reflect the cover area of the algorithm to some extent. Especially when the GDOP value is less than 10, we think the system can meet the primary positioning demand. Fig. 10 shows the proposed method's distribution of CRLB and GDOP.

E. LED's Inclination Angle Test

To gain a more intuitive grasp of the influence of the LED's inclination angle, a real test is performed to show its influence in ranging when we use the traditional calibrated-ranging-model-based method. The PD was deployed at different distances, and we changed its inclination angle. A ranging model is calibrated in advance, and the calibrated model and

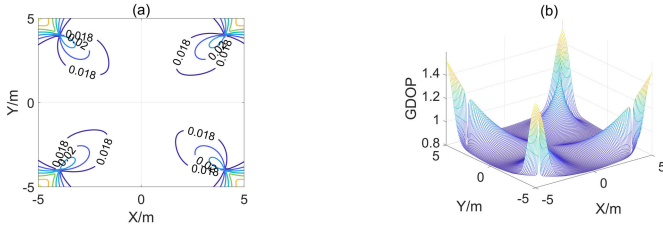


Fig. 10. Theoretical performance of the proposed method. (a) CRLB distribution of the proposed method. (b) GDOP distribution of the proposed method.

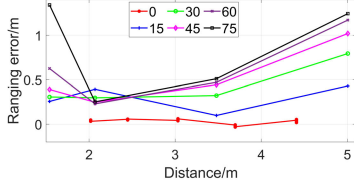


Fig. 11. Localization error in the different distances when the PD has different inclination angles.

RSS are used to obtain the distance. The inclination angle is set to 0°, 15°, 30°, 45°, 60°, and 75°, respectively. The distribution of the ranging error is shown in Fig. 11. It is obvious that when the inclination angle is 0°, all the ranging errors are smaller than 10 cm. However, as the inclination angle increased, the ranging errors also increased. The mean ranging errors of different inclination angles are calculated, and we use these mean errors as the threshold of the random ranging error in the following comparison experiments.

F. Localization Experiments and Analysis

1) *Accuracy Evaluation Indicator*: The mean absolute error is the main indicator we used for results' accuracy evaluation in this article, representing the difference between the estimated value and the ground truth

$$\text{MAE} = \|\mathbf{eva} - \mathbf{gt}\|_2 \quad (22)$$

where $\mathbf{eva}$ represents the estimated vector, $\mathbf{gt}$ is the ground truth, and $\|\cdot\|_2$ is the two-norm.

Besides, we pay much more attention to analyzing the distribution of the errors. Thus, the cdf of the errors is always presented via the figures.

2) *Localization Experiments*: Here, we chose two typical tracks to test the performance of the proposed method: a square track and a two-loop circle track. Moreover, some static points are added to increase the complexity of the trajectory.

There are some comparative experiment sets: one is algorithms' comparison, including the traditional multiple-anchor-based method, the proposed method without weight, and the proposed method with weight. The traditional multiple-anchor-based method used a single-LED-based anchor, and the anchors were deployed at the same position as the dual-LED anchors. Besides, we added a random error in the distance. The magnitude of the random error is introduced above. The other set is the inclination angle comparison. To simplify the experiment, the PD kept a fixed inclination angle in each experiment rather than changed with the movements.

Fig. 12 shows the calculated trajectories in the square track experiment by different methods, and Fig. 13 shows the

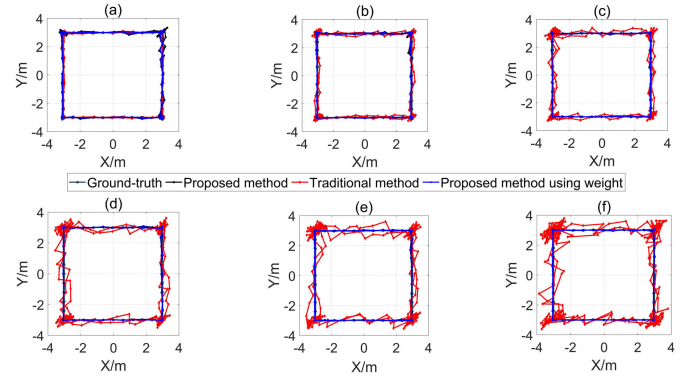


Fig. 12. Localization results of a square track when the PD has different inclination angles. (a) 0°. (b) 15°. (c) 30°. (d) 45°. (e) 60°. (f) 75°.

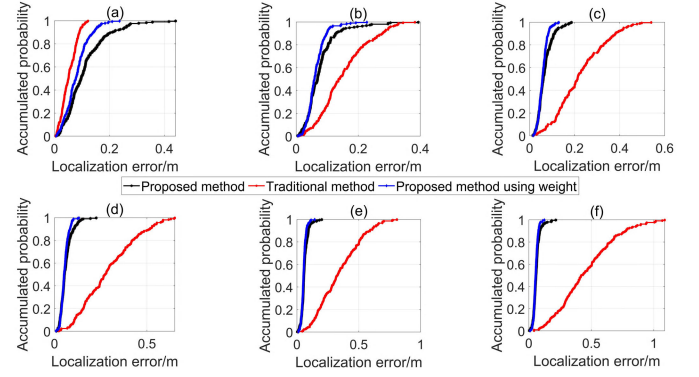


Fig. 13. Localization errors' cdf of a square track when the PD has different inclination angles. (a) 0°. (b) 15°. (c) 30°. (d) 45°. (e) 60°. (f) 75°.

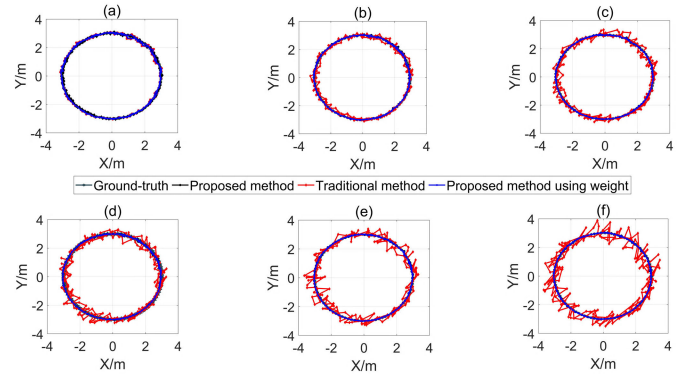


Fig. 14. Localization results of a circle track when the PD has different inclination angles. (a) 0°. (b) 15°. (c) 30°. (d) 45°. (e) 60°. (f) 75°.

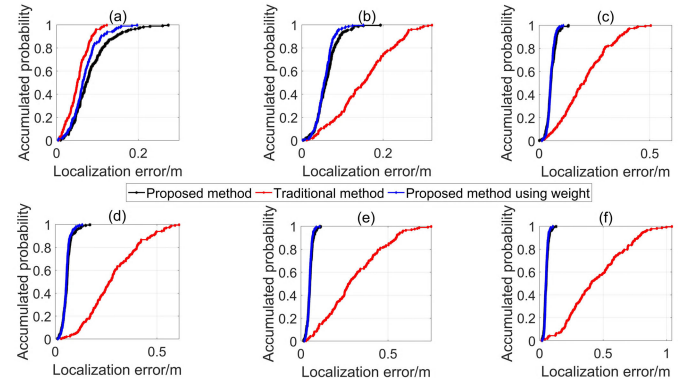


Fig. 15. Localization errors' cdf of a circle track when the PD has different inclination angles. (a) 0°. (b) 15°. (c) 30°. (d) 45°. (e) 60°. (f) 75°.

localization errors' cdf. In Fig. 12, it is easy to find that the trajectory obtained by the traditional multiple-anchor-based

TABLE I
STATISTICAL RESULTS OF DIFFERENT METHODS' LOCALIZATION ERRORS WITH DIFFERENT INCLINATION ANGLES

Method	Track	Inclination angle	Mean/m	Max/m	Min/m	STD/m
Traditional multiple-anchors-based method	Square track	0°	0.053	0.122	0.004	0.028
		15°	0.153	0.383	0.009	0.080
		30°	0.225	0.540	0.018	0.111
		45°	0.291	0.651	0.024	0.148
		60°	0.343	0.808	0.011	0.172
		75°	0.457	1.087	0.042	0.228
	Circle track	0°	0.053	0.123	0.005	0.026
		15°	0.152	0.320	0.012	0.073
		30°	0.208	0.505	0.012	0.109
		45°	0.283	0.609	0.029	0.129
		60°	0.304	0.748	0.029	0.166
		75°	0.446	1.042	0.016	0.230
The proposed method with weight	Square track	0°	0.081	0.237	0.006	0.043
		15°	0.064	0.228	0.006	0.031
		30°	0.058	0.130	0.016	0.020
		45°	0.055	0.132	0.008	0.020
		60°	0.055	0.144	0.012	0.019
		75°	0.056	0.125	0.011	0.017
	Circle track	0°	0.069	0.197	0.004	0.033
		15°	0.056	0.151	0.002	0.021
		30°	0.053	0.109	0.013	0.015
		45°	0.055	0.132	0.013	0.018
		60°	0.050	0.106	0.021	0.013
		75°	0.052	0.105	0.020	0.013
The proposed method without weight	Square track	0°	0.111	0.441	0.004	0.075
		15°	0.078	0.393	0.005	0.054
		30°	0.068	0.186	0.014	0.033
		45°	0.062	0.226	0.011	0.032
		60°	0.061	0.201	0.003	0.028
		75°	0.061	0.216	0.003	0.029
	Circle track	0°	0.083	0.274	0.010	0.047
		15°	0.062	0.194	0.005	0.028
		30°	0.055	0.132	0.001	0.020
		45°	0.058	0.170	0.011	0.023
		60°	0.052	0.113	0.015	0.017
		75°	0.053	0.130	0.018	0.017

TABLE II
STATISTICAL RESULTS OF OUR PROPOSED METHOD'S DISTANCE ERRORS WITH DIFFERENT INCLINATION ANGLES

Track	Inclination angle	Mean/m	Max/m	Min/m	STD/m
Square track	0°	0.048	0.122	0.012	0.021
	15°	0.036	0.112	0.010	0.015
	30°	0.032	0.069	0.010	0.010
	45°	0.031	0.068	0.012	0.010
	60°	0.030	0.076	0.007	0.009
	75°	0.030	0.066	0.014	0.008
Circle track	0°	0.040	0.088	0.011	0.014
	15°	0.033	0.069	0.017	0.009
	30°	0.030	0.052	0.010	0.007
	45°	0.030	0.060	0.014	0.008
	60°	0.028	0.050	0.014	0.006
	75°	0.029	0.051	0.011	0.006

method is rougher and rougher with increasing PD's inclination angle. As long as the inclination angle is not zero, big errors start to appear. On the contrary, the trajectory calculated by the proposed method always keeps a high accuracy, which is almost the same as the ground truth. It is clear that the improvement of the proposed method

with weight is not very obvious compared with the proposed method without weight, and their coinciding degree is very high. However, there are still some jump points at the inflection point when the no-weight method is used. Obviously, the weight-matrix-based method solves this problem well, which can also be obtained from Fig. 13. Similarly, in Fig. 13,

TABLE III
COMPARISON WITH PREVIOUS WORKS

Ref	Method	Performance	Solution
[40]	RSS (Trilateration)	Medium error >1.5m in different scenes tests.	Simulation, a $\pm 20^\circ$ inclination angle is added.
[41]	RSS	Minimum error > 1m in different signal noise ratio tests.	Simulation.
[42]	RSS	Median errors of 12.7 cm and maximal errors of 21.1 cm.	Simulation and the real test, requires the PD to keep horizontal.
[21]	TDOA	Mean error: mm-level.	Simulation without any error, frequency exchange will bring problems.
[43]	TDOA	Mean error: 0.403m.	Simulation with a 1ns-inherent-clock-error.
[24]	Fingerprint (RSS)	About 0.2m or even bigger (different hardware parameters).	Simulation.
Ours	Phase-difference-based method (LS)	Mean error: about 5.5cm in different tests.	Simulation with time synchronization error and receiver noise.

with the inclination angle increase, the localization accuracy decreases rapidly if the traditional multiple-anchor-based method is used. The maximum error is more than 1 m when the inclination angle is 75° .

Fig. 14 shows the calculated trajectories in the circle track experiment by different methods, and Fig. 15 shows the localization errors' cdf. It can obtain the same conclusion as the square track experiment.

Besides, we counted the localization errors of different methods and the proposed method's ranging errors. Table I shows the statistical results of the localization errors, and Table II shows the statistical results of the ranging errors. From Table I, many conclusions can be obtained as follows: If the traditional multiple-anchor-based method is used, it can achieve high accuracy when the inclination angle is zero, reaching about 5 cm. However, with the inclination angle increase, the localization accuracy decreases rapidly. The mean error can reach half a meter, and the maximum error is even more than 1 m when the inclination angle is 75° . Significantly, the proposed method solves this problem well. Whatever the inclination changes, the localization accuracy keeps a centimeter level, which is about 5–10 cm. Nevertheless, the maximum error of some inclination angles can still reach 50 cm, which means there are still some unacceptable errors, and it is obviously quite big for some applications. The distance-determined weight matrix is imported to decrease the big error's value further. Although the improvement of the proposed method with weight is not very obvious in localization accuracy, which is similar to the proposed method without weight and is also about 5–10 cm, it shows excellent potential in handling the big error. The maximum error is kept under 20 cm, which can also be observed in Figs. 13 and 15. However, the proposed method with weight did not show a great advantage in handling small errors compared with the method without weight. They have little difference in small error handling performance. From Table II, the ranging accuracy is the main area of concern. Whatever the inclination angle changes, it is always less than 5 cm. Here, the time synchronization error and receiver noise are all taken into consideration.

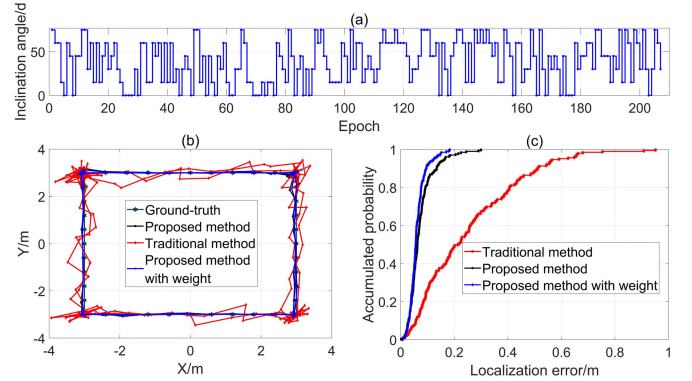


Fig. 16. Results when the inclination angle is always changing in the square track. (a) Inclination angle distribution. (b) Localization results of different methods. (c) Localization errors' cdf.

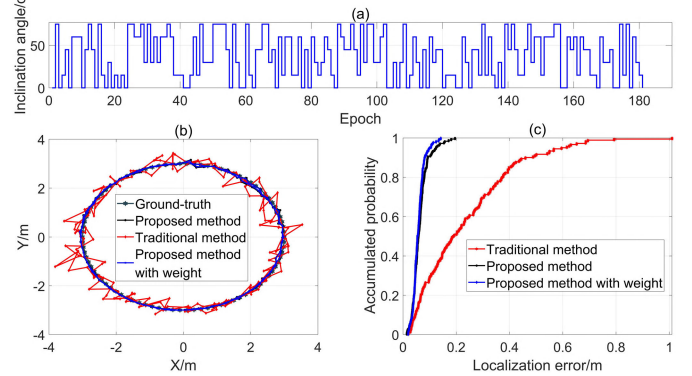


Fig. 17. Results when the inclination angle is always changing in the circle track. (a) Inclination angle distribution. (b) Localization results of different methods. (c) Localization errors' cdf.

Moreover, another experiment is performed to discuss further the proposed method's ability to eliminate the influence of PD's inclination angle (two kinds of trajectories are selected). The PD's inclination angle is changed randomly in the whole trajectory. Figs. 16(a) and 17(a) show the inclination angle at each epoch, and Figs. 16(b) and 17(b) show the localization results of the different methods. The cdf of localization errors is presented in Figs. 16(c) and 17(c). It is clear that the inclination angle does not affect the

proposed method, and about 95% of errors are less than 10 cm. On the contrary, the influence of the inclination angle on the traditional method is relatively severe, and the maximum error can be above 1 m, and the mean localization error can also reach several decimeters.

3) *Comparison With Previous Work*: To further analyze the proposed method's performance, we compare it with some similar positioning systems developed in recent years. Some papers [21], [24], [40], [41], [42], [43] are chosen to compare with ours, which use a similar method or similar experimental setup to ours. A detailed description is shown in Table III.

One is compared with the RSS-based methods and can be further divided into trilateration and fingerprinting. The former always uses a signal strength attenuation model to calculate the distances and the crossing location-based localization method. This method-based system's results are not good enough due to the requirements of the experimental setups. The real tests' accuracy can always be above 1 m, and the simulation tests' accuracy just reaches a decimeter level. Of course, some of them can be a centimeter-level accuracy because of their near-perfect simulation setups. The latter is seldom studied in high accurate IPSs, like VLP, because its accuracy is heavily connected to the fingerprint points' density. Thus, its accuracy always is a decimeter level or even higher.

The other is compared with the methods based on TDOA, which always uses the phase difference to calculate the distance or the distance difference. These methods are similar to ours and most use simulation for performance evaluation. Due to the experiment setups, the performance is significantly different. Some can be a millimeter level, but some just reach a decimeter level, even if a tiny error is added.

IV. CONCLUSION

This article proposed a dual-LED-based VLP system, and a phase-difference-based ranging method was introduced to replace the traditional calibrated-ranging-model-based methods. First, a dual-LED anchor integrated by two closely spaced LEDs was elaborately designed to replace the traditional separated LEDs for ranging, which dramatically eliminated the workload of the ranging model's calibration and decreased the dependence on frequency exchange. The main design included the works of LEDs' interval design and frequency assignment. Second, the ranging method based on the phase difference was proposed to solve the issues faced by the traditional methods, including the influence of hardware heterogeneity, environmental differences, and receiver placement. Finally, a weight matrix was used for final localization, which showed great potential in handling big localization errors. A set of experiments were performed to evaluate the performance of the proposed method, which fully considers the time synchronization error and receiver noise. The results indicated that it obtained a 5–10-cm localization accuracy and showed more advantages than the traditional empirical-ranging-model-based method, especially when the PD's inclination angle was not zero. In cases where the weight matrix was used, the maximum error was kept under 20 cm, which shows superiority in many indoor applications.

There are still many works that can be done in the future. The first is that some actual experiments should be performed

to evaluate the proposed method. The second task is to use multisource data for integrated positioning, such as lidar and visible light.

REFERENCES

- [1] Y. Li et al., "Toward location-enabled IoT (LE-IoT): IoT positioning techniques, error sources, and error mitigation," *IEEE Internet Things J.*, vol. 8, no. 6, pp. 4035–4062, Mar. 2021.
- [2] N. El-Sheimy and Y. Li, "Indoor navigation: State of the art and future trends," *Satell. Navigat.*, vol. 2, no. 1, p. 7, Dec. 2021.
- [3] X. Cao, Y. Zhuang, X. Yang, X. Sun, and X. Wang, "A universal Wi-Fi fingerprint localization method based on machine learning and sample differences," *Satell. Navigat.*, vol. 2, no. 1, p. 27, Dec. 2021, doi: [10.1186/s43020-021-00058-8](https://doi.org/10.1186/s43020-021-00058-8).
- [4] X. Yang, D. Chen, J. Huai, X. Cao, and Y. Zhuang, "An improved wireless positioning algorithm based on the LSTM network," in *Proc. China Satell. Navigat. Conf. (CSNC)*, C. Yang and J. Xie, Eds. Singapore: Springer, 2021, pp. 616–627.
- [5] Z. Liu et al., "Precise, low-cost and large-scale indoor positioning system based on audio dual-chirp signals," *IEEE Trans. Veh. Technol.*, vol. PP, pp. 1–10, 2022.
- [6] Y. Zhuang, C. Zhang, J. Huai, Y. Li, L. Chen, and R. Chen, "Bluetooth localization technology: Principles, applications, and future trends," *IEEE Internet Things J.*, vol. 9, no. 23, pp. 23506–23524, Dec. 2022.
- [7] A. R. J. Ruiz and F. S. Granja, "Comparing ubisense, bespoon, and decawave UWB location systems: Indoor performance analysis," *IEEE Trans. Instrum. Meas.*, vol. 66, no. 8, pp. 2106–2117, Aug. 2017.
- [8] N. Chaudhary, L. N. Alves, and Z. Ghassemloooy, "Current trends on visible light positioning techniques," in *Proc. 2nd West Asian Colloq. Opt. Wireless Commun. (WACOWC)*, Apr. 2019, pp. 100–105.
- [9] Y. Li et al., "Inertial sensing meets machine learning: Opportunity or challenge?" *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 9995–10011, Aug. 2022.
- [10] X. Wang, G. Chen, X. Cao, Z. Zhang, M. Yang, and S. Jin, "Robust and accurate step counting based on motion mode recognition for pedestrian indoor positioning using a smartphone," *IEEE Sensors J.*, vol. 22, no. 6, pp. 4893–4907, Mar. 2022.
- [11] R. Kusche, S. O. Schmidt, and H. Hellbruck, "Indoor positioning via artificial magnetic fields," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.
- [12] M. Shu, G. Chen, and Z. Zhang, "Efficient image-based indoor localization with MEMS aid on the mobile device," *ISPRS J. Photogramm. Remote Sens.*, vol. 185, pp. 85–110, Mar. 2022.
- [13] F. Seguel, P. Palacios-Jativa, C. A. Azurdia-Meza, N. Krommenacker, P. Charpentier, and I. Soto, "Underground mine positioning: A review," *IEEE Sensors J.*, vol. 22, no. 6, pp. 4755–4771, Mar. 2022.
- [14] Y. Zhuang et al., "A survey of positioning systems using visible LED lights," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 3, pp. 1963–1988, 3rd Quart., 2018.
- [15] M. Maheepala, A. Z. Kouzani, and M. A. Joordens, "Light-based indoor positioning systems: A review," *IEEE Sensors J.*, vol. 20, no. 8, pp. 3971–3995, Apr. 2020.
- [16] S. H. Yang, E. M. Jeong, D. R. Kim, H. S. Kim, Y. H. Son, and S. K. Han, "Indoor three-dimensional location estimation based on led visible light communication," *Electron. Lett.*, vol. 49, no. 1, pp. 54–55, 2013.
- [17] C. Danakis, M. Afgani, G. Povey, I. Underwood, and H. Haas, "Using a CMOS camera sensor for visible light communication," in *Proc. IEEE Globecom Workshops*, Dec. 2012, pp. 1244–1248.
- [18] M.-G. Moon, S.-I. Choi, J. Park, and J. Y. Kim, "Indoor positioning system using LED lights and a dual image sensor," *J. Opt. Soc. Korea*, vol. 19, no. 6, pp. 586–591, Dec. 2015.
- [19] H. H. H. Huang, L. F. L. Feng, G. N. G. Ni, and A. Yang, "Indoor imaging visible light positioning with sampled sparse light source and mobile device," *Chin. Opt. Lett.*, vol. 14, no. 9, pp. 90602–90606, 2016.
- [20] Y. Wang, B. Hussain, and C. P. Yue, "Arbitrarily tilted receiver camera correction and partially blocked LED image compensation for indoor visible light positioning," *IEEE Sensors J.*, vol. 22, no. 6, pp. 4800–4807, Mar. 2022.
- [21] S.-Y. Jung, S. Hann, and C.-S. Park, "TDOA-based optical wireless indoor localization using LED ceiling lamps," *IEEE Trans. Consum. Electron.*, vol. 57, no. 4, pp. 1592–1597, Nov. 2011.

- [22] Z. Zhu, B. Zhu, Y. Wang, Z. Shi, and Y. Jiang, "Angle-of-arrival estimator for light signals based on optimized photodiode array," in *Proc. Int. Conf. Comput., Netw. Commun. (ICNC)*, Feb. 2019, pp. 62–66.
- [23] P. Chen, M. Pang, D. Che, Y. Yin, D. Hu, and S. Gao, "A survey on visible light positioning from software algorithms to hardware," *Wireless Commun. Mobile Comput.*, vol. 2021, pp. 1–20, Feb. 2021.
- [24] L. Hua, Y. Zhuang, F. Gu, L. Qi, and J. Yang, "AdVLP: Unsupervised visible light positioning by adversarial deep learning," *Meas. Sci. Technol.*, vol. 32, no. 6, Jun. 2021, Art. no. 064003.
- [25] X. Sun et al., "RSS-based visible light positioning using nonlinear optimization," *IEEE Internet Things J.*, vol. 9, no. 15, pp. 14137–14150, Aug. 2022.
- [26] J. Kunthoth, A. Karkar, S. Al-Maadeed, and A. Al-Ali, "Indoor positioning and wayfinding systems: A survey," *Human-Centric Comput. Inf. Sci.*, vol. 10, no. 1, pp. 1–41, Dec. 2020.
- [27] H. Q. Tran and C. Ha, "Machine learning in indoor visible light positioning systems: A review," *Neurocomputing*, vol. 491, pp. 117–131, Jun. 2022.
- [28] J. Lian, Z. Vatansever, M. Noshad, and M. Brandt-Pearce, "Indoor visible light communications, networking, and applications," *J. Phys., Photon.*, vol. 1, no. 1, Jan. 2019, Art. no. 012001.
- [29] S. Zhang, K. Liu, Y. Zhang, and Y. Wang, "A coarse fingerprint-assisted multiple target indoor device-free localization with visible light sensing," *IEEE Sensors J.*, vol. 22, no. 2, pp. 1461–1473, Nov. 2021.
- [30] A. H. A. Bakar, T. Glass, H. Y. Tee, F. Alam, and M. Legg, "Accurate visible light positioning using multiple-photodiode receiver and machine learning," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–12, 2021.
- [31] C. Guo, "Implementation and improvement of Bayes's theorem for indoor visible light positioning system," *Opt. Eng.*, vol. 58, no. 2, Feb. 2019, Art. no. 026110.
- [32] M. Saadi, Z. Saeed, T. Ahmad, M. K. Saleem, and L. Wuttisittikulkij, "Visible light-based indoor localization using k-means clustering and linear regression," *Trans. Emerg. Telecommun. Technol.*, vol. 30, no. 2, Feb. 2019, Art. no. e3480.
- [33] X. Zhu, W. Qu, T. Qiu, L. Zhao, M. Atiquzzaman, and D. O. Wu, "Indoor intelligent fingerprint-based localization: Principles, approaches and challenges," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 4, pp. 2634–2657, 4th Quart., 2020.
- [34] B. Zhu, J. Cheng, Y. Wang, J. Yan, and J. Wang, "Three-dimensional VLC positioning based on angle difference of arrival with arbitrary tilting angle of receiver," *IEEE J. Sel. Areas Commun.*, vol. 36, no. 1, pp. 8–22, Jan. 2018.
- [35] L. Hua, Y. Zhuang, L. Qi, J. Yang, and L. Shi, "Noise analysis and modeling in visible light communication using Allan variance," *IEEE Access*, vol. 6, pp. 74320–74327, 2018.
- [36] B. Cao, S. Wang, S. Ge, and W. Liu, "Improving positioning accuracy of UWB in complicated underground NLOS scenario using calibration, VBUKF, and WCA," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–13, 2021.
- [37] S. Cao, X. Chen, X. Zhang, and X. Chen, "Effective audio signal arrival time detection algorithm for realization of robust acoustic indoor positioning," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 10, pp. 7341–7352, Oct. 2020.
- [38] Y. Sun, F. Zhang, and Q. Wan, "Wireless sensor network-based localization method using TDOA measurements in MPR," *IEEE Sensors J.*, vol. 19, no. 10, pp. 3741–3750, Jan. 2019.
- [39] L. Santoro, D. Brunelli, and D. Fontanelli, "On-line optimal ranging sensor deployment for robotic exploration," *IEEE Sensors J.*, vol. 22, no. 6, pp. 5417–5426, Mar. 2022.
- [40] L. Li and F. Zhao, "Epsilon: A visible light based positioning system," in *Proc. 11th USENIX Symp. Netw. Syst. Design Implement. (NSDI)*, Apr. 2014, pp. 1–14. [Online]. Available: <https://www.microsoft.com/en-us/research/publication/epsilon-a-visible-light-based-positioning-system/>
- [41] Q. Wang et al., "Light positioning: A high-accuracy visible light indoor positioning system based on attitude identification and propagation model," *Int. J. Distrib. Sensor Netw.*, vol. 14, no. 2, Feb. 2018, Art. no. 155014771875826.
- [42] D. Plets et al., "Three-dimensional visible light positioning: An experimental assessment of the importance of the LEDs' locations," in *Proc. Int. Conf. Indoor Positioning Indoor Navigat. (IPIN)*, Sep. 2019, pp. 1–6.
- [43] C. Zhang, D. Li, X. Feng, L. Tan, L. Yang, and X. Yang, "Indoor visible light positioning method based on TDOA and fingerprint," in *Proc. IEEE Int. Conf. Electron. Technol., Commun. Inf. (ICETCI)*, Aug. 2021, pp. 397–401.



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